Signal iOS 8.36 (build 1981)

Source-to-binary verification of the App Store build

Prepared by commandersaki with Claude (Opus 5.5, Anthropic) in Claude Code · 8 November 2026

Result. The compiled code (`__TEXT,__text`) of all 21 binaries in the App Store app is **byte-identical** to a local build of tag `8.36.0.1981` with Xcode 27.1 (27A9275). That covers Signal's five own binaries and all 16 bundled frameworks. Every other difference is one already explained for earlier versions: the signature-size field in every binary; the UUID and the linker's GOT-slot order in `SignalServiceKit` alone, with every stub resolving to the same target; build metadata in plists; and the app icons, which are rasterised at build time and differ by at most 1/255.

1 Method

This is the same method as for 8.35.1, which describes it in full.

1. The source is cloned at the Xcode Cloud path `/Volumes/workspace/repository`.
2. `make dependencies` fetches the dependencies.
3. An unsigned `xcodebuild archive` runs with the scheme's App Store Release configuration, using the exact Xcode build recorded in the shipped `Info.plist`.
4. `verify/compare_build.sh` and `verify/explain_residuals.sh` compare the result with the decrypted App Store IPA.

Input	Value
App Store app	Signal 8.36 (1981), downloaded from the App Store (version ID `894738344`) and decrypted with unfairplay. The encrypted IPA passes all 26,778 of its code-signature page hashes. Decryption changed nothing outside the FairPlay-encrypted ranges apart from the `cryptid` flag.
Source	Tag `8.36.0.1981`, commit `012f2f2`, the commit recorded in the shipped `Info.plist`. Pods submodule `fae00ed`; libsignal `0.106.1` (Swift Package Manager) and RingRTC `v2.73.0`, at the versions the tag pins.
Toolchain	Xcode 27.1 (27A9275), matching the shipped `DTXcodeBuild`.

2 Results

Check	Result
`__TEXT,__text`, all 21 binaries	Byte-identical (68.2 MB, hashed with Apple's `segedit` and `shasum`)
Functions, Signal's 5 binaries	150,344 of 150,344 match after address normalisation
Strings, symbols, linked libraries	Identical
Header bytes	`__LINKEDIT` size (signature) in all 21 binaries; `LC_UUID` in `SignalServiceKit` only
GOT slot order	`SignalServiceKit` only (999 + 1 stub bytes). All 1,873 + 999 stubs resolve to the same target.
Plists	38 differ, only in `BuildDetails` time, `BuildMachineOSBuild`, `CFBundleVersion` and `DTAppStoreToolsBuild`

Check	Result
Asset catalogs	All renditions identical (2,832 main, 2,542 share extension, 8 NSE), except the 72 rasterised app-icon bitmaps, which `actool` renames randomly and which differ slightly in content
Loose icons	107 of 14,400 and 176 of 23,104 pixels differ, by at most 1/255 in one channel

3 Limitations

As for earlier versions:

- The public source itself is not reviewed.
- Prebuilt dependencies (libsignal's Rust library, RingRTC, WebRTC, LibMobileCoin) are checked against their pinned versions, not rebuilt from their own source.
- The causes of the UUID and icon-rasterisation differences are inferred, not proven.

4 Reproducing

```
git clone https://github.com/signalapp/Signal-iOS.git /Volumes/workspace/repository
cd /Volumes/workspace/repository && git checkout 8.36.0.1981 && make dependencies
# xcodebuild archive with Xcode 27.1 (27A9275), as in the 8.35.1 report, then:
verify/compare_build.sh signal-ios-8.36-1981.decrypted.ipa <built Signal.app>
versions/8.36.0.1981/out/ipa_vs_build
verify/explain_residuals.sh signal-ios-8.36-1981.decrypted.ipa <built Signal.app>
versions/8.36.0.1981/out/residuals
```

The decrypted IPA is attached to the `signal-8.36` release. The app is also in `versions/8.36.0.1981/signal_dec/`, and all outputs are in `versions/8.36.0.1981/out/`.